multiple features to align with similar activation directions, supporting disentangled and interpretable representations.

Motivated by the superposition hypothesis (Elhage et al., 2022), SAEs assume that features combine *additively* in the decoder, so reconstruction is linear in $\boldsymbol{z}$. This corresponds to a strong form of the linear representation hypothesis applied to decoding, which has recently been questioned (Engels et al., 2025). When multiple features co-activate, their joint effect may not be well captured by a linear sum—for example, a "coffee" feature and a "star" feature may require a reconstruction direction distinct from either individual atom to capture the "Starbucks" concept. This motivates extending the decoder to explicitly model feature interactions, while preserving a linear and interpretable encoder.

3.2. Design Principles for Feature Interactions

We extend sparse autoencoders to capture higher-order feature interactions by establishing design principles grounded in prior work. Each architectural choice in PolySAE follows directly from these principles.

P1. Linear Encoding (Interpretability). Each sparse code coefficient z_i is derived by a *linear projection* of the input activation $\boldsymbol{x}$. The linear representation hypothesis in mechanistic interpretability posits that learned features should correspond to *directions* in activation space (Elhage et al., 2022; Bricken et al., 2023), a view supported by the success of linear probes for extracting semantic content (Belinkov, 2022; Alain & Bengio, 2016).

P2. Polynomial Reconstruction (Expressivity). The decoder may capture compositional structure by using polynomial terms in $\boldsymbol{z}$. Modeling *compositional* structure, *i.e.* how features interact, polynomials have a strong precedent in the literature: Volterra series (Volterra, 1959) represent nonlinear systems as sums of multilinear kernels, second-order pooling (Carreira et al., 2012; Gao et al., 2016) captures feature co-occurrences via outer products, and polynomial networks (Chrysos et al., 2022b) parameterize functions as products of linear projections.

P3. Factorized Interaction Structure (Coherence & Efficiency). Higher-order terms should operate in a low-dimensional subspace aligned with the linear feature space. Using a shared projection $\boldsymbol{U}$ ensures that interactions are compositions of the same underlying features. This alignment principle underlies factorized interaction models (Rendle, 2010; Blondel et al., 2016) and compact bilinear pooling (Gao et al., 2016; Kim et al., 2017). Constraining interactions to low-rank subspaces imposes a strong inductive bias, favoring coherent, reusable interaction modes over arbitrary pairwise composition.

P4. Structural Constraints (Parsimony & Identifiability). Lower-order terms should have higher representational capacity than higher-order terms, following polynomial approximation theory (Mason & Handscomb, 2002). The latent interaction subspace should have orthonormal columns to ensure geometrically distinct directions. Orthogonality constraints are standard in dictionary learning and independent component analysis to prevent degenerate solutions (Arora et al., 2015; Bao et al., 2016; Hyvärinen & Oja, 2000). Orthonormality removes rotational ambiguity and ensures the model does not allocate redundant capacity to correlated interaction directions.

3.3. PolySAE: Polynomial Sparse Autoencoder

To satisfy **P1** PolySAE adopts the standard SAE encoder (Huben et al., 2024; Bricken et al., 2023) to first performs a linear map followed by sparsification:

$$\boldsymbol{z} = \mathcal{S}\big(\mathrm{ReLU}(\boldsymbol{E}^\top \boldsymbol{x} + \boldsymbol{b}_{\mathrm{enc}})\big), \qquad \boldsymbol{z} \in \mathbb{R}^{d_{\mathrm{sae}}}, \tag{1}$$

where feature i activates when $\boldsymbol{x}$ aligns with direction $\boldsymbol{e}_i$, enabling visualization, clustering, and causal intervention via activation patching (Meng et al., 2022).

Following **P2**, we extend the decoder to include quadratic and cubic terms:

$$\hat{\boldsymbol{x}} = \boldsymbol{b}_{\mathrm{dec}} + \boldsymbol{y}_1 + \lambda_2\, \boldsymbol{y}_2 + \lambda_3\, \boldsymbol{y}_3, \tag{2}$$

where $\boldsymbol{y}_1 = \boldsymbol{A}\,\boldsymbol{z}$, $\boldsymbol{y}_2 = \boldsymbol{B}\,(\boldsymbol{z} \otimes \boldsymbol{z})$, $\boldsymbol{y}_3 = \boldsymbol{\Gamma}\,(\boldsymbol{z} \otimes \boldsymbol{z} \otimes \boldsymbol{z})$, and $\lambda_2, \lambda_3 \in \mathbb{R}$ are learnable scalar coefficients that control the contribution of each polynomial order. Setting $\lambda_2 = \lambda_3 = 0$ recovers a standard linear sparse autoencoder, making PolySAE a strict generalization of existing SAE architectures. This can be viewed as a third-order Volterra expansion (Volterra, 1959) or a Π-*net* polynomial parameterization (Chrysos et al., 2022b), adapted to sparse codes.

However, explicitly modeling all pairwise or higher-order feature combinations would require $O(d_{\mathrm{sae}}^2)$ or $O(d_{\mathrm{sae}}^3)$ parameters, leading to unstructured interaction effects and a high risk of overfitting. Following **P3** and **P4**, we constrain interactions to a low-rank subspace:

$$\begin{aligned}
\boldsymbol{y}_1 &= (\boldsymbol{z}\,\boldsymbol{U})\,\boldsymbol{C}^{(1)\top}, \\
\boldsymbol{y}_2 &= \big((\boldsymbol{z}\,\boldsymbol{U}_{:,1:R_2}) * (\boldsymbol{z}\,\boldsymbol{U}_{:,1:R_2})\big)\,\boldsymbol{C}^{(2)\top}, \\
\boldsymbol{y}_3 &= \big((\boldsymbol{z}\,\boldsymbol{U}_{:,1:R_3}) * (\boldsymbol{z}\,\boldsymbol{U}_{:,1:R_3}) * (\boldsymbol{z}\,\boldsymbol{U}_{:,1:R_3})\big)\,\boldsymbol{C}^{(3)\top},
\end{aligned} \tag{3}$$

where $*$ denotes element-wise product and $\boldsymbol{C}^{(k)} \in \mathbb{R}^{d \times R_k}$ are output projection matrices. This parameterization restricts the interaction dictionaries to rank at most R_k, enforcing a strong inductive bias on how features may combine.

Notice that this parameterization satisfies **P3** by applying a single projection $\boldsymbol{U}$ to the sparse code and forming interactions via polynomial operations: $\boldsymbol{z}\boldsymbol{U}$, $(\boldsymbol{z}\boldsymbol{U}) * (\boldsymbol{z}\boldsymbol{U})$, and

$(\boldsymbol{zU}) * (\boldsymbol{zU}) * (\boldsymbol{zU})$. Using the same projected representation at every order ensures that interaction effects remain aligned with the linear feature basis and interpretable as compositions of the same underlying features.

Furthermore, PolySAE satisfies the parsimony aspect of **P4** by following nested low-rank approximation (Grasedyck et al., 2013) and utilizing ranks (R_1, R_2, R_3) with $R_1 \geq R_2 \geq R_3$. This nested structure means $\mathrm{span}(\boldsymbol{U}_{:,1:R_3}) \subset \mathrm{span}(\boldsymbol{U}_{:,1:R_2}) \subset \mathrm{span}(\boldsymbol{U})$. In practice, $R_2 = R_3 \ll R_1$ (*e.g.*, $R_2 = R_3 = 64$) suffices to capture most interaction structure, confirming our hypothesis that higher-order contributions are low-dimensional (Section 4).

Finally, to satisfy the identifiability aspect of **P4**, we enforce orthonormality of the interaction subspace. Following Stiefel optimization (Absil et al., 2008; Bonnabel, 2013), we impose $\boldsymbol{U}^\top \boldsymbol{U} = \boldsymbol{I}$ via QR retraction after each gradient step. We use positive QR retraction (Edelman et al., 1998), which corrects column signs to ensure continuity and avoids discontinuous representation changes during training.

3.4. Discussion

Context-Dependent Dictionary Structure. In standard SAEs, each feature i is associated with a fixed dictionary atom $\boldsymbol{d}_i$: regardless of context, activating feature i contributes $z_i \boldsymbol{d}_i$ to the reconstruction. PolySAE fundamentally alters this picture. Because reconstruction includes higher-order terms, the effective contribution of a feature becomes *context-dependent*, varying with which other features are simultaneously active.

This can be seen by expanding Equation (2). The linear term defines a dictionary over individual features, while the quadratic and cubic terms define dictionaries over feature pairs and triples, respectively. Under our low-rank factorization, these dictionaries are implicitly given by

$$\begin{aligned}
\boldsymbol{A} &= \boldsymbol{C}^{(1)} \boldsymbol{U}^\top && \in \mathbb{R}^{d \times d_{\text{sae}}}, \\
\boldsymbol{B} &= \boldsymbol{C}^{(2)} (\boldsymbol{U}_{:,1:R_2} \odot \boldsymbol{U}_{:,1:R_2})^\top && \in \mathbb{R}^{d \times d_{\text{sae}}^2}, \\
\boldsymbol{\Gamma} &= \boldsymbol{C}^{(3)} (\boldsymbol{U}_{:,1:R_3} \odot \boldsymbol{U}_{:,1:R_3} \odot \boldsymbol{U}_{:,1:R_3})^\top && \in \mathbb{R}^{d \times d_{\text{sae}}^3},
\end{aligned} \tag{4}$$

where $\boldsymbol{A}$ is the *linear dictionary*, $\boldsymbol{B}$ the *pairwise interaction dictionary*, and $\boldsymbol{\Gamma}$ the *triple interaction dictionary*. Column (i, j) of $\boldsymbol{B}$ specifies how the co-activation $z_i z_j$ modifies the reconstruction, while column (i, j, k) of $\boldsymbol{\Gamma}$ specifies the contribution arising from the joint activation $z_i z_j z_k$. The computational form in Equation (3) is algebraically equivalent to Equation (4) but avoids explicitly materializing the d_{sae}^2- and d_{sae}^3-dimensional dictionaries.

Compositional Capacity. Using the same d_{sae} base features as a standard SAE, PolySAE can support interaction-driven structure across $\binom{d_{\text{sae}}}{2} \cdot R_2 + \binom{d_{\text{sae}}}{3} \cdot R_3$ feature pairs and triples, enabling a substantially larger space of distinct semantic compositions without increasing the number of learned features. This capacity is mediated through a shared low-rank interaction space: rather than allocating independent parameters to each feature combination, interactions are expressed via R_2 and R_3 shared modes. As a result, a large number of potential feature combinations are realized through a small number of reusable interaction directions, reflecting the empirically observed low-dimensional structure of feature interactions.

Parameter Efficiency. PolySAE modifies only the decoder; the encoder is unchanged. A standard SAE has $2d\, d_{\text{sae}} + d + d_{\text{sae}}$ parameters. When the linear term is full rank ($R_1 = d$), PolySAE adds $\Delta P = d^2 + d(R_2 + R_3) + 2$ parameters. With the empirically optimal choice $R_2 = R_3$ and $R_2 \in [0.06R_1,\ 0.11R_1]$, this yields $\Delta P = (1.12\text{–}1.22)\, d^2$ (up to constants). For GPT-2 small ($d = 768$, $d_{\text{sae}} = 16{,}384$), this corresponds to an increase of $\sim$2.5–3% of the full SAE.

4. Empirical Evaluation

4.1. Experimental Setup

Our training pipeline is built by extending `SAELens` (Bloom et al., 2024) to include PolySAE. We train and evaluate our methods against the standard SAE with **three sparsification strategies**, TopK (Gao et al., 2025), BatchTopK (Bussmann et al., 2024), and Matryoshka (Bussmann et al., 2025b). Throughout all experiments, we use a sparsity level of $K = 64$ with 16,384 latents trained on residual-stream activations from **four pretrained language models of different scales**: Gemma-2-2B (Gemma Team, 2024) (layer 19), Pythia-410M and Pythia-1.4B (Biderman et al., 2023) (layers 15 and 12, respectively), and GPT-2 Small (Radford et al., 2019) (layer 8). Training uses 500M tokens (300M for GPT-2 Small) with context length 128. For Gemma-2-2B and GPT-2 Small, we use OpenWebText (Gokaslan et al., 2019); for Pythia models, we use an uncopyrighted variant of the deduplicated Pile (Gao et al., 2021). We evaluate learned features using SAEBench (Karvonen et al., 2025), which reports reconstruction metrics on held-out data from the training distribution and sparse probing performance on **six classification tasks**: Bias in Bios (De-Arteaga et al., 2019), AG News (Zhang et al., 2015), EuroParl (Koehn, 2005), GitHub programming languages (CodeParrot, 2022), Amazon Sentiment, and Amazon-15 (Hou et al., 2024). For more implementation details see Section B.

4.2. Reconstruction and Semantic Modeling

We evaluate models along two axes: *(Q1) reconstruction fidelity* and *(Q2) semantic modeling of the learned representations*. Reconstruction quality is measured using mean squared error between the decoder output and the unnormalized network activations. To assess semantic structure, we

Table 1. F1 Scores (%) across datasets at K=1. Format: F1 / Wasserstein ($\times 10^{-3}$). Mean Probing column shows mean F1 across datasets.

LLM	SAE variant	MSE	Mean F1	Europarl	Bios	Amazon Sentiment	GitHub	AG News	Amazon 15
GPT-2 Small	Topk	**0.52**	67.1	67.7 / 19.0	61.0 / 7.7	76.0 / 4.3	63.4 / 8.7	71.4 / 8.5	63.3 / 2.8
	Topk + PolySAE	0.55	**77.9**	**86.1 / 35.2**	**75.5 / 16.8**	**83.1 / 9.7**	**73.0 / 20.6**	**81.0 / 18.9**	**69.0 / 6.7**
	BTopk	**0.53**	65.7	67.4 / 19.0	59.6 / 7.3	68.8 / 4.4	68.1 / 8.8	65.3 / 8.2	65.1 / 2.9
	BTopk + PolySAE	0.54	**78.0**	**92.0 / 39.9**	**70.9 / 17.3**	**84.2 / 8.5**	**74.4 / 18.5**	**83.2 / 20.0**	**63.2 / 6.0**
	Matryoshka	0.60	65.7	65.8 / 12.5	61.2 / 4.0	76.2 / 3.2	60.9 / 7.9	68.1 / 4.3	62.1 / 2.4
	Matr. + PolySAE	**0.58**	**77.7**	**95.0 / 30.0**	**72.9 / 14.3**	**81.4 / 8.1**	**71.5 / 18.6**	**77.4 / 16.0**	**68.0 / 5.6**
Pythia-410m	Topk	**0.03**	71.2	96.1 / 2.0	67.4 / 1.1	61.5 / 0.7	64.6 / 1.5	71.8 / 1.2	65.9 / 0.4
	Topk + PolySAE	0.04	**77.0**	**96.7 / 6.8**	**70.8 / 3.8**	**75.9 / 2.3**	**74.0 / 5.3**	**73.3 / 4.0**	**71.5 / 1.4**
	BTopk	**0.03**	65.0	90.9 / 0.8	60.5 / 0.3	63.9 / 0.4	59.7 / 1.1	58.7 / 0.3	56.6 / 0.3
	BTopk + PolySAE	0.04	**77.3**	**97.8 / 8.2**	**74.0 / 4.1**	**74.6 / 2.1**	**75.2 / 4.8**	**78.6 / 4.4**	**63.5 / 1.3**
	Matryoshka	**0.04**	64.2	79.1 / 0.6	63.6 / 0.3	64.4 / 0.4	62.3 / 1.1	58.7 / 0.3	57.3 / 0.3
	Matr. + PolySAE	**0.04**	**74.6**	**99.2 / 2.8**	**71.0 / 1.2**	**66.9 / 1.3**	**81.8 / 3.7**	**64.8 / 1.2**	**63.8 / 0.9**
Pythia-1.4b	Topk	**0.23**	75.9	**97.8** / 1.6	72.5 / 1.4	69.5 / 0.8	69.3 / 1.9	77.3 / 1.4	69.0 / 0.5
	Topk + PolySAE	**0.23**	**81.9**	96.8 / **7.9**	**77.2 / 6.4**	**88.1 / 3.7**	**74.7 / 9.2**	**83.4 / 6.3**	**71.1 / 2.4**
	BTopk	**0.22**	64.6	74.0 / 0.6	65.0 / 0.5	57.2 / 0.6	63.3 / 2.3	65.2 / 0.4	63.2 / 0.4
	BTopk + PolySAE	0.23	**76.4**	**93.7 / 4.5**	**73.0 / 3.4**	**67.1 / 3.1**	**73.8 / 8.2**	**77.6 / 3.4**	**73.2 / 2.1**
	Matryoshka	0.24	64.4	70.2 / 0.5	62.8 / 0.5	**63.1** / 0.6	65.6 / 1.9	64.1 / 0.4	60.8 / 0.4
	Matr. + PolySAE	**0.23**	**72.1**	**91.1 / 2.9**	**72.4 / 2.0**	58.0 / **2.1**	**68.2 / 6.7**	**73.6 / 2.1**	**69.4 / 1.5**
Gemma2-2b	Topk	**1.59**	67.7	78.6 / 5.3	**69.6** / 7.1	**71.8** / 4.4	60.7 / 6.1	60.7 / 7.2	64.8 / 2.7
	Topk + PolySAE	1.65	**68.4**	**86.8 / 12.0**	64.7 / **16.8**	64.5 / **10.5**	**64.1 / 16.1**	**61.9 / 16.9**	**68.5 / 6.3**
	BTopk	**1.58**	64.8	68.3 / 1.9	67.6 / 2.6	**71.1** / 2.8	64.4 / 4.5	59.9 / 2.7	57.6 / 1.9
	BTopk + PolySAE	1.68	**69.4**	**92.8 / 13.2**	**78.3 / 18.3**	56.4 / **10.2**	**65.0 / 16.1**	**64.0 / 18.8**	**60.0 / 6.4**
	Matryoshka	1.69	60.9	60.8 / 0.7	64.0 / 0.8	57.3 / 1.5	61.5 / 2.5	**61.0** / 0.8	60.5 / 1.0
	Matr. + PolySAE	**1.64**	**65.6**	**77.6 / 2.1**	**67.5 / 3.1**	**61.7 / 4.9**	**63.7 / 8.8**	60.9 / **3.5**	**62.2 / 3.3**

use two complementary metrics.

Probing. We evaluate the linear separability of semantic concepts in the learned sparse representations by training logistic regression classifiers on SAE activations to predict ground-truth labels across multiple datasets. For each task, classification is performed using the feature with the largest mean activation difference between positive and negative classes, isolating semantic signal at the feature level.

Distributional separation. Probing relies on post-hoc decision boundaries and may not fully reflect the intrinsic geometry of the representation. We therefore additionally compute the 1-Wasserstein distance between class-conditional activation distributions. Unlike probing, which evaluates separability at a specific threshold, the Wasserstein distance captures global distributional separation, with larger values indicating more distinct semantic separation across space.

Table 1 demonstrates that **across four language models and three sparsification strategies**, PolySAE achieves *comparable MSE to standard SAE* across all configurations, confirming that polynomial decoding does not sacrifice reconstruction fidelity. For probing, PolySAE **consistently outperforms SAE by large margins** with mean gains of more than 10% on GPT-2, and 8% on average across models (Pythia-410M, Pythia-1.4B, and Gemma2-2B) and sparsifiers. Crucially, PolySAE also achieves **consistently substantially higher Wasserstein distances**, with improvements of approximately 2–10$\times$ across all other models.

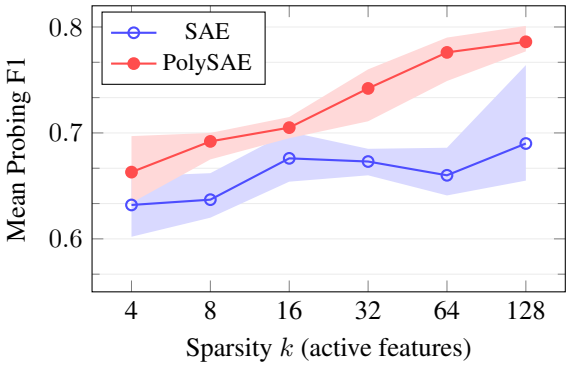

Figure 3. **Probing Mean F1 vs. sparsity** k. Shaded regions show range across widths (2k–16k). PolySAE consistently outperforms SAE with significant separation at higher k.

This indicates that the gains observed in probing accuracy reflect *genuinely better-separated class-conditional representations*, rather than improvements driven solely by favorable decision boundaries.

4.3. Ablations

PolySAE Enables Competitive Performance with Sparser Codes. We ask *(Q3) whether PolySAE's capacity to model feature interactions enables the use of sparser representations*. Figure 3 shows probing F1 as a function of active features k, with shaded regions indicating variance across dictionary widths (2k–16k). PolySAE consistently